DRUGS · HEAD AND NECK ONCOLOGY

Methotrexate

Methotrexate is an antifolate. In head and neck cancer it is an older, well-tolerated palliative drug for recurrent disease. In ENT inflammatory disease it is a steroid-sparing immunosuppressant used in granulomatosis with polyangiitis, autoimmune inner ear disease and other vasculitides. It is given once weekly, and prescribing it daily by mistake has killed patients.

1 CHAPTER ONE At a glance

The drug on one page.

1.1 Summary

METHOTREXATE

	SUMMARY
Class	Antifolate antimetabolite and immunosuppressant
How it works	Inhibits dihydrofolate reductase (cytotoxic doses); at low doses, increases adenosine release and inhibits T-cell activation (anti-inflammatory)
ENT uses	Palliative treatment of recurrent or metastatic HNSCC (weekly intravenous); maintenance of remission in non-severe granulomatosis with polyangiitis; autoimmune inner ear disease (steroid-sparing, limited evidence); sarcoidosis
Dose	Palliative 40 mg/m2 intravenously weekly; immunosuppression commonly 15 to 25 mg orally or subcutaneously **once weekly**, with folic acid (typically 5 mg weekly on another day)
Monitoring	Full blood count, liver and renal function every 2 weeks at first, then every 2 to 3 months
Avoid	Pregnancy (teratogenic; stop 3 to 6 months before conception), significant renal or liver impairment, active infection

2 CHAPTER TWO Using it in ENT

Practice.

2.1 Granulomatosis with polyangiitis

After induction with steroids and rituximab or cyclophosphamide, maintenance is often with rituximab, azathioprine or methotrexate; limited sinonasal disease may be treated with methotrexate and steroids. ENT management of crusting, septal perforation, saddle nose and subglottic stenosis continues alongside.

3 CHAPTER THREE Safety

What goes wrong.

3.1 Adverse effects

Mucositis, myelosuppression, hepatotoxicity and fibrosis, pneumonitis (cough and breathlessness: stop and investigate), nausea, infection risk, teratogenicity.

3.2 Interactions

Trimethoprim and co-trimoxazole (severe marrow toxicity), NSAIDs, penicillins and PPIs (reduced excretion), other hepatotoxic drugs and alcohol.

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Once-weekly** dosing and folic acid.
- The co-trimoxazole interaction.
- Its role in GPA.

4.2 Common pitfalls

COMMON PITFALLS

- Prescribing methotrexate daily.
- Adding co-trimoxazole for a sinus infection in a patient on methotrexate.

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